

Sunaina Yaseen

AI Agentic Engineer • Full-Stack AI Developer • Builder of Autonomous Systems

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PROFESSIONAL SUMMARY

Driven AI Agentic Engineer who turns ambitious ideas into working, autonomous systems. Built **GenSpark Builds** — an end-to-end, AI-powered custom PC platform driven by autonomous planning agents, tool-calling workflows, real-time compatibility reasoning, vendor orchestration, and YOLOv8-based visual QA. Energized by hard problems at the intersection of **multi-agent architecture, RAG pipelines, and full-stack engineering**, with hands-on command of React.js, Flask, LangChain, LangGraph, and the OpenAI/Gemini APIs. Thrives on designing intelligent systems that don't just respond — they plan, reason, and act.

TECHNICAL SKILLS

Programming Languages: Python, JavaScript

Frontend: React.js, HTML5, CSS3, Bootstrap

Backend: Flask, REST APIs

Agentic AI: LangChain, LangGraph, Multi-Agent Systems, Tool Calling, AI Planning, Memory, MCP (Model Context Protocol), Prompt Engineering

LLMs: OpenAI API, Gemini API

AI / ML: YOLOv8 (Ultralytics), OpenCV, RAG (Retrieval-Augmented Generation), FAISS, ChromaDB

Database: MySQL

Deployment & Tools: Vercel, Railway, Git, GitHub, Postman

PROJECTS

GenSpark Builds — Agentic AI Platform for Custom PC Building & Assembly

Final Year Project

GitHub: github.com/Sunainayaseen

- **Architected** a modular multi-agent system — dedicated agents for requirement analysis, PC recommendation, budget optimization, compatibility validation, vendor selection, QA verification, and notifications — replacing a single monolithic AI workflow with a scalable, specialized pipeline.
- **Engineered** autonomous LangGraph planning workflows that execute multi-step tasks end-to-end (analyze → recommend → validate → optimize → finalize) with minimal user intervention.
- **Built** a LangChain tool-calling agent that independently invokes component search, compatibility-checking, price-calculation, and vendor-finder tools to fulfill user requests from start to finish.
- **Designed** a rule-based compatibility engine paired with a self-correction agent that automatically detects failed compatibility checks and re-validates replacement components on the fly.
- **Grounded** component recommendations in reality by integrating Retrieval-Augmented Generation (FAISS/ChromaDB) over live MySQL inventory spanning 90+ hardware components across 12 categories — eliminating model guesswork.
- **Orchestrated** an AI-driven vendor network connecting 20+ verified vendors, automating order assignment and full lifecycle tracking across 4 user roles (Customer, Vendor, Admin, Rider).
- **Deployed** a YOLOv8-powered QA agent that validates vendor-uploaded assembly photos against ordered components before shipment approval — catching errors before they reach the customer.
- **Delivered** 30+ REST APIs powering both AI agents and web clients, complete with live rider tracking for order delivery.
- **Shipped** the entire platform end-to-end — React.js frontend, Flask/Python backend, MySQL database — live in production on Vercel and Railway.

Tech Stack: React.js, Flask/Python, MySQL, OpenAI/Gemini API, LangChain, LangGraph, MCP, FAISS/ChromaDB, YOLOv8, Git/GitHub

YOLOv8 Hardware Component Detection

Personal / Academic Project

- **Built** a real-time computer vision system that detects and classifies computer hardware parts (RAM, GPU, CPU, motherboard) live from a camera feed.
- **Powered** detection with YOLOv8 (Ultralytics) for high-speed object detection and precise bounding-box localization.

- **Applied** OpenCV for real-time video capture and frame processing, streamlining hardware inventory identification.

Learning Management System (LMS)

Academic Project

- **Built** a full-stack LMS with separate student and teacher roles, each with tailored dashboards and permissions.
- **Implemented** course management, assignment submission, a quiz system, and student progress tracking.

Tech Stack: *React.js, Flask, MySQL*

Hospital Management System

Academic Project

- **Developed** a hospital management platform covering patient records, doctor appointment scheduling, and prescription management.
- **Built** a billing module and admin panel to manage end-to-end hospital operations.

Tech Stack: *React.js, Flask, MySQL*

ACHIEVEMENTS

- **Built** and shipped GenSpark Builds, an end-to-end Agentic AI platform, as Final Year Project.
- **Designed** autonomous multi-agent workflows from the ground up using LangGraph and LangChain.
- **Automated** quality assurance with a custom computer-vision pipeline powered by YOLOv8.
- **Delivered** four full-stack applications — GenSpark Builds, LMS, Hospital Management System, and YOLOv8 detection — using React & Flask.

EDUCATION

BS Computer Science (BSCS)

2026

University of Education, Lahore

CERTIFICATIONS & TRAINING

- Full Stack Web Development — Ideoversity Training Institute